

TrES-1b

R-1505

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_220610 · created 2026-08-01T15:20:35+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459737.822 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.053
Transit depth (Rp/Rs)^2	3.11 +/- 1.86 [%]
Orbital Inclination (inc)	86.29 +/- 1.43
Airmass coefficient 1 (a1)	0.36 +/- 0.16
Airmass coefficient 2 (a2)	0.68 +/- 0.66
Scatter in the residuals of the lightcurve fit is	43.8 %
Best Comparison Star	#3 - [554, 468]
Optimal Aperture	4.7
Optimal Annulus	7.76
Transit Duration (day)	0.0134 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.053 vs 0.1358 (deviation 0.76σ)
Duration fit vs published	0.0134 d vs 0.1042 d (deviation 10.81σ)
Mid-time O–C	-4357.7 min
Depth SNR	3.3
Residual scatter	43.8 %

Light curve

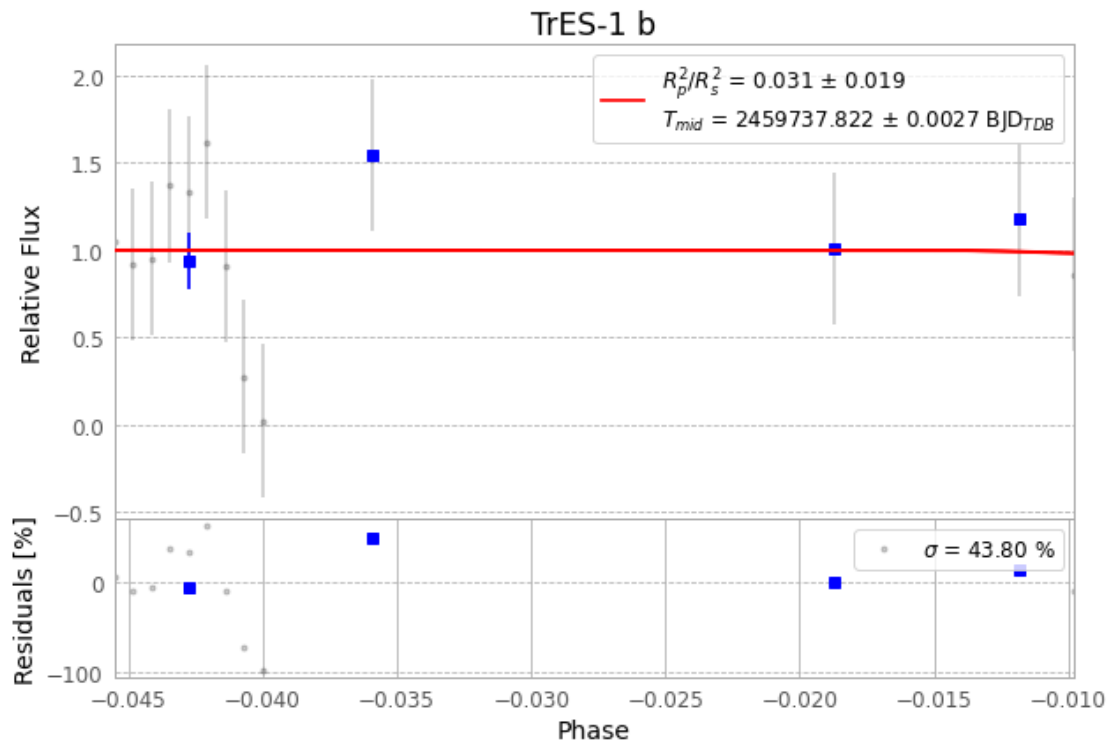


Figure 1 — FinalLightCurve_TrES-1 b_2022-06-09.png (SHA-256 0ab8549666623ae6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [34, 428], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 506b15c21d4c991b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

111 FITS frames (73 MB), TRES-1220610050313.FITS → TRES-1220610103313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-220610.log (fae3a0d8e241...), FinalLightCurve_TrES-1 b_2022-06-09.png (0ab854966662...), FinalLightCurve_TrES-1 b_2022-06-09.csv (4b712de8d407...)

Compute resources

Wall time 215.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 15:20Z.